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Critical Convergence Theorem for a Dimensionless Invariant in Confined Wave Systems[*]

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We establish a critical convergence theorem for a dimensionless invariant characterizing confined wave systems reaching a saturated stationary regime. Within a general variational framework, we prove that any normalizable stationary solution of a coercive energy functional converges, in the thermodynamic limit, toward a universal ratio between stored energy and transport energy, determined solely by the asymptotic dilation structure of the system. This abstract invariant admits a direct physical realization in the measurable form $\mathcal{I} = V f^3/v^3$, relating confinement volume, characteristic frequency, and phase velocity.

I. INTRODUCTION

Physical wave systems never occur as ideal plane waves. They are necessarily confined, normalizable, and coupled to a finite-energy substrate. This fact imposes strong structural constraints on the stationary regimes accessible to real physical systems.

We show here that any confined wave system admitting a variational description and reaching a *saturated* stationary regime necessarily selects a universal, dimensionless ratio between stored energy and transport energy. This ratio is not postulated: it emerges as a fixed point of the dilation symmetry of the energy functional in the thermodynamic limit.

Claimed novelty and formal robustness. Unlike classical virial-type theorems, which relate energetic contributions within specific dynamical models, the present result establishes the existence of a universal dimensionless invariant selected solely by asymptotic dilation structure and normalization, independently of any specific force, potential, or geometry, and provides for the first time a directly measurable physical realization in the form $\mathcal{I} = V f^3/v^3$.

II. VARIATIONAL FRAMEWORK

Let $\Omega_L \subset \mathbb{R}^d$ be a bounded domain and $\mathcal{H}_L = H_0^1(\Omega_L)$. We consider a coercive energy functional

$$\mathcal{E}_L[u] = \mathcal{E}_L^{(\mathrm{pot})}[u] + \mathcal{E}_L^{(\mathrm{kin})}[u], \tag{1}$$

where $\mathcal{E}_L^{(\mathrm{pot})}[u] \geq 0$ represents stored (localized) energy and $\mathcal{E}_L^{(\mathrm{kin})}[u] \geq 0$ the energy associated with transport.

Stationary states are obtained as constrained extrema of $\mathcal{E}_L$ under the normalization condition

$$\|u\|_{L^2(\Omega_L)} = 1. \tag{2}$$

Remark 1 (Status of $\mathcal{E}_L^{(\mathrm{kin})}$). In this work, $\mathcal{E}_L^{(\mathrm{kin})}$ denotes the energetic contribution associated with *transport* (or flux) of the confined mode, in a broad sense: it may correspond, depending on the physical context, to kinetic, radiative (Poynting-type), or interfacial energy. No specific microscopic form is assumed beyond coercivity and asymptotic homogeneity.

III. CRITICAL CONVERGENCE THEOREM

Theorem 1 (Critical convergence). *Let $\Omega_L \uparrow \mathbb{R}^d$ be a regular exhaustion. Under the unitary dilation $u^\lambda(x) = \lambda^{d/2}u(\lambda x)$, assume that the directional derivatives satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathcal{E}_L^{(\mathrm{pot})}[u^\lambda]\bigg|_{\lambda=1} = p\,\mathcal{E}_L^{(\mathrm{pot})}[u] + o_L(1), \tag{3}$$

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathcal{E}_L^{(\mathrm{kin})}[u^\lambda]\bigg|_{\lambda=1} = -q\,\mathcal{E}_L^{(\mathrm{kin})}[u] + o_L(1), \tag{4}$$

with $p, q > 0$. Assume moreover that the remainder terms $o_L(1)$ are uniform for λ in a neighborhood of 1. If u_L is a stationary solution under the constraint (2), then

$$\mathcal{J}_L \equiv \frac{\mathcal{E}_L^{(\mathrm{pot})}[u_L]}{\mathcal{E}_L^{(\mathrm{kin})}[u_L]} \xrightarrow[L\to\infty]{} \mathcal{J}_* = \frac{q}{p}, \tag{5}$$

independently of the geometry of Ω_L and microscopic details.

Remark 2 (Scope of the term "universal"). Here, "universal" means independent of microscopic details and confinement geometry *within* the class of coercive functionals admitting dilation homogeneity in the sense of Eqs. (3)–(4). No claim beyond this class is made.

IV. PROOF

Proof. Consider the constrained functional $\mathcal{L}_L[u] = \mathcal{E}_L[u] - \mu_L\big(\|u\|^2_{L^2(\Omega_L)} - 1\big)$. Stationarity of u_L implies $\mathrm{d}\mathcal{L}_L[u_L^\lambda]/\mathrm{d}\lambda|_{\lambda=1} = 0$. The dilation path $u_L^\lambda(x) =$

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$\lambda^{d/2}u_L(\lambda x)$ preserves the L^2 norm, thus cancelling the constraint term. We obtain

$$0 = \frac{\mathrm{d}}{\mathrm{d}\lambda}\mathcal{E}_L^{(\mathrm{pot})}[u_L^\lambda]\Big|_{\lambda=1} + \frac{\mathrm{d}}{\mathrm{d}\lambda}\mathcal{E}_L^{(\mathrm{kin})}[u_L^\lambda]\Big|_{\lambda=1}.$$

Using Eqs. (3) and (4), we find

$$p\,\mathcal{E}_L^{(\mathrm{pot})}[u_L] - q\,\mathcal{E}_L^{(\mathrm{kin})}[u_L] = o_L(1).$$

Coercivity ensures $\mathcal{E}_L^{(\mathrm{kin})}[u_L] > 0$, hence

$$\mathcal{J}_L = \frac{q}{p} + o_L(1).$$

Taking the limit $L \to \infty$ completes the proof. □

V. PHYSICAL REALIZATION OF THE INVARIANT

Definition 1 (Measurable invariant)**.** For a confined and stable wave mode with effective volume V, characteristic frequency f, and phase velocity v, we define

$$\mathcal{I} = \frac{V f^3}{v^3}. \tag{6}$$

Remark 3 (Status of the identification $\mathcal{J}_* \mapsto \mathcal{I}$). The identification $\mathcal{I} = Vf^3/v^3$ is understood here as an *operational projection* of the abstract variational fixed point $\mathcal{J}_*$ onto directly measurable macroscopic quantities, rather than a consequence derived solely from the variational formalism.

Remark 4. The cubic exponent in the definition of $\mathcal{I}$ is imposed by scale invariance in three spatial dimensions, ensuring that $\mathcal{I}$ is a dimensionless quantity independent of local metric details.

Remark 5 (Physical origin of the cubic law). In a saturated stationary regime, transport energy is controlled by a volumetric flux characterized by the phase velocity v, while stored energy is governed by the density of accessible modes within the confined volume. In three spatial dimensions, this density scales as f^3. The ratio $\mathcal{I} = Vf^3/v^3$ thus emerges as the natural physical realization of the abstract variational fixed point $\mathcal{J}_*$, independently of any specific microscopic interaction.

VI. ILLUSTRATIVE APPLICATION

Remark 6 (Interpretation of phase velocity). In the microgravity regime considered here, the bubble itself constitutes the confined wave mode of the interfacial system. Its migration velocity therefore corresponds to the effective phase velocity of the mode, not to a distinct group velocity.

Example 1. For a gas bubble in microgravity with radius $R \simeq 10^{-3}$ m, the volume is $V \simeq 4.2\times10^{-9}\,\mathrm{m}^3$. With $f \simeq 2\,\mathrm{Hz}$ and $v \simeq 1.3\times10^{-3}\,\mathrm{m\,s^{-1}}$, we obtain, by direct application of Eq. (6),

$$\mathcal{I} \simeq \frac{(4.2\times10^{-9})\,2^3}{(1.3\times10^{-3})^3} \simeq 1.7\times10^{-2}.$$

VII. DISCUSSION

The invariant $\mathcal{I}$ is absent from plane-wave-based descriptions, as such models explicitly ignore confinement and global normalization. Its emergence thus provides a structural criterion distinguishing idealized mathematical solutions from physically realizable systems. The notions of coercivity and energetic stability invoked here fall within the classical framework of functional analysis and the stability of matter [1, 2].

Appendix A: Axiomatic formulation

Theorem 1 relies solely on the following assumptions, which define a general class of confined wave systems accessible to a variational description.

- **A1 (Coercivity).** The energy functional $\mathcal{E}_L$ is bounded from below on $\mathcal{H}_L$.
- **A2 (Energetic decomposition).** The functional decomposes as $\mathcal{E}_L = \mathcal{E}_L^{(\mathrm{pot})} + \mathcal{E}_L^{(\mathrm{kin})}$ with $\mathcal{E}_L^{(\mathrm{pot})}, \mathcal{E}_L^{(\mathrm{kin})} \geq 0$.
- **A3 (Asymptotic dilation homogeneity).** Under unitary dilation, $\mathcal{E}_L^{(\mathrm{pot})}$ and $\mathcal{E}_L^{(\mathrm{kin})}$ satisfy Eqs. (3)–(4) up to negligible boundary terms.
- **A4 (Thermodynamic limit).** Boundary contributions and remainder terms $o_L(1)$ vanish as $L \to \infty$.
- **A5 (Saturation).** For sufficiently large L, at least one nontrivial stationary state exists.
- **A6 (Normalizability).** Admissible physical states satisfy

  $$u_L \in H_0^1(\Omega_L), \qquad \|u_L\|_{L^2(\Omega_L)} = 1,$$

  excluding idealized plane waves and ensuring finite, localizable energy.

Remark 7. These axioms constitute sufficient *structural conditions* to establish critical convergence of $\mathcal{J}_L$ toward the universal fixed point $\mathcal{J}_*$, without invoking any additional assumptions on microscopic interactions or operator structure.

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